Theory: the statistics behind the report

The Perks of Being a First Mover · CS215 Open Project · Group 02

This document explains every statistical method in our final report, why it fits the question, and how to compute it, with a small worked example for each. The examples use small made-up numbers so they can be checked by hand. The real data has 160 million games.

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1. Definitions

1.1 Symbols

Symbol	Meaning
R_i^{White}, R_i^{Black}	The two players' Lichess ratings in game i.
d_i	Rating gap in game i: $d_i = R_i^{\text{White}} - R_i^{\text{Black}}$ (positive when White is higher rated).
S_i	White's score in game i: 1 for a win, $\frac{1}{2}$ for a draw, 0 for a loss.
$E(d)$	Expected score of a player rated d points above the opponent (the Elo formula).
h	The **true** first-move advantage: how many rating points moving first is worth, for the games we are describing (all of Lichess, one opening, one speed, …). One fixed number that nobody knows.
$\hat{h}$	Our **estimate** of h, worked out from the games we actually have ("h-hat"). The hat always means "estimated from data".
$\text{SE}(\hat{h})$	Standard error: how much $\hat{h}$ would typically change if we repeated the study with a fresh batch of games (Section 4).
95% CI	Confidence interval $\hat{h} \pm 1.96\,\text{SE}$: a range that contains the true h in 95% of repeated studies.
n_d	Number of games with rating gap d.
K	$\ln 10/400 \approx 0.005757$, which turns $10^{x/400}$ into e^{Kx}.
$I(\hat{h})$	Fisher information: how sharply the log-likelihood peaks at $\hat{h}$. $\text{SE} = 1/\sqrt{I}$.
z	An estimate divided by its standard error. Close to standard normal if the true value is 0.
p	p-value: the chance of a $\lvert z \rvert$ at least this big if the true value were 0.

1.2 h and $\hat{h}$: why a single number has a distribution

Once we have our games, $\hat{h}$ is just one number: put the games in, get (say) 28.2 out. So what does it mean to talk about the *distribution* of $\hat{h}$, its standard error, or a confidence interval?

The point is that the games themselves are partly luck: a blunder here, a lost-on-time there. Our games are one random sample of all the games that could have been played. A different month, or a different 100 games, would give a slightly different $\hat{h}$. So:

- h is **fixed** but unknown: a property of how chess is played on Lichess.
- $\hat{h}$ is **random before we collect the data**, because it's computed from random games. After we collect them it's one number, but it's one draw from that randomness.

- The **distribution of** $\hat{h}$ (its *sampling distribution*) describes the values we *would* get if we repeated the whole study many times with fresh games. Its centre is the true h, and its spread is the standard error.

A coin example. Flip a coin 100 times and get 54 heads. Your estimate of the chance of heads is $\hat{p} = 0.54$, one number. But you know another 100 flips might give 49 or 57: the estimate has a spread of about $\sqrt{0.5 \times 0.5/100} = 0.05$, even though the coin's true chance p never changes. $\hat{h}$ is exactly like $\hat{p}$: the 0.54 is our one run, and the spread describes the other runs we could have had.

Where the spread comes from. The likelihood (Section 3) scores every possible value of h by how well it explains our games. If the peak is *sharp*, only values very close to $\hat{h}$ explain the data, so another batch of games would give nearly the same estimate: a small spread. If the peak is *flat*, many values fit almost equally well, so another batch could land far away: a large spread. The Fisher information I measures how sharp the peak is, and $\text{SE} = 1/\sqrt{I}$. A standard theorem says that with many games,

$$\hat{h} \;\approx\; \mathcal{N}\big(h,\, \text{SE}^2\big),$$

a bell curve centred on the true h.

Why it matters for us. With 100 games, SE ≈ 35 points (Section 4), so another 100 games could easily give -10 or $+60$. With 160 million games it's a few hundredths of a point, so any other month would give almost the same $\hat{h}$. And it's what lets us compare: saying "the aggressive openings have a larger h" only means something if the gap between the two $\hat{h}$ values is bigger than this luck-of-the-draw spread. That's what the $\pm$ and the z-tests check.

In the rest of this document, and in the report, numbers like "$h = +11.9 \pm 0.1$" are estimates: strictly they are $\hat{h}$, with their 95% CI.

2. The model: White's advantage in rating points

2.1 The Elo expected score

Lichess ratings follow the Elo idea: if player A is rated $d = R_A - R_B$ points above player B, A's expected score (win = 1, draw = $\frac{1}{2}$, loss = 0) is

$$E(d) = \frac{1}{1 + 10^{-d/400}}$$

$E(0) = 0.5$, $E(50) = 0.5715$, $E(-50) = 0.4285$. Every 400 points multiply the odds of winning by 10.

2.2 Why we don't just compare win rates

A raw White score mixes the first move with the rating gap. Suppose a 1540 plays a 1570 with White and wins. The ratings alone expect White to score $E(-30) = 0.4569$, so this win is $1 - 0.4569 = 0.543$ points "better than expected". A win between two 1570s is only 0.5 better than expected. Comparing every result to what the ratings predict makes all games comparable, whatever the gap.

2.3 Adding a White bonus h

We model the first move as a bonus of h rating points for White:

$$E_i(h) = \frac{1}{1 + 10^{-(d_i + h)/400}}, \qquad d_i = R_i^{\text{White}} - R_i^{\text{Black}}$$

h is the quantity we estimate. Measuring it in rating points means "White is worth about h points" can be compared across ratings, speeds and openings. Draws are counted as half a point, and we report draw rates separately.

3. Estimating h: maximum likelihood

3.1 The likelihood

Treat each game's score S_i as an outcome with mean $E_i(h)$. The (Bernoulli) log-likelihood is

$$\ell(h) = \sum_i \Big[S_i \ln E_i(h) + (1 - S_i) \ln \big(1 - E_i(h)\big) \Big]$$

With draws ($S_i = \frac{1}{2}$) this is a *quasi*-likelihood: it isn't the exact probability of the data, but it is maximised at the same kind of estimate and is the standard way to handle fractional scores.

3.2 Setting the derivative to zero

Write $K = \ln 10/400$, so $E_i = 1/(1 + e^{-K(d_i+h)})$. The logistic function has the handy derivative

$$\frac{dE_i}{dh} = K\, E_i(1 - E_i)$$

Differentiating ℓ and simplifying:

$$\ell'(h) = \sum_i \left[\frac{S_i}{E_i} - \frac{1 - S_i}{1 - E_i} \right] K E_i(1 - E_i) = K \sum_i \big(S_i - E_i(h) \big)$$

So the maximum-likelihood estimate $\hat{h}$ is the value where

$$\boxed{\sum_i E_i(\hat{h}) = \sum_i S_i}$$

that is, *White's expected total score equals White's actual total score*. This is exactly the intuition: choose the bonus that makes the model agree with what happened.

3.3 Solving it: Newton's method

There is no closed form, so we iterate. The second derivative is

$$\ell''(h) = -K^2 \sum_i E_i(1 - E_i) \;<\; 0,$$

so ℓ is concave and has a single maximum. Newton's method moves to where a straight-line approximation of ℓ' crosses zero:

$$h_{\text{new}} = h - \frac{\ell'(h)}{\ell''(h)} = h + \frac{K \sum_i (S_i - E_i)}{K^2 \sum_i E_i(1 - E_i)}$$

Games with the same rating gap d share the same E, so we only need, for each gap d, the number of games n_d and White's total score S_d. That is why our program stores counts by rating gap rather than individual games.

Worked example: estimating h from 100 games

Three groups of games, by rating gap d:

Gap d	Games n_d	White's score S_d	Expected at $h=0$: $n_d E(d)$
0	40	22	$40 \times 0.5000 = 20.00$
$+50$	30	19	$30 \times 0.5715 = 17.15$
-50	30	13	$30 \times 0.4285 = 12.85$
Total	100	54	50.00

White scored 54 where the ratings expect 50, so $h > 0$. Here $K = \ln 10/400 = 0.0057565$.
Step 1 ($h = 0$):

$$\begin{aligned} \ell' &= K\,(54 - 50) = 0.023026, \\ -\ell'' &= K^2 \textstyle\sum n_d E_d (1 - E_d) = K^2\big[40(0.25) + 2 \times 30(0.5715)(0.4285)\big] \\ &= K^2 \times 24.693 = 0.00081827, \\ \text{step} &= 0.023026/0.00081827 = 28.14, \quad \text{giving } h = 28.14 \end{aligned}$$

Step 2 ($h = 28.14$): the expected scores are now 0.5404, 0.6106 and 0.4686 per game, adding up to 53.992. $\ell' = K(54 - 53.992) = 0.000048$, and $-\ell'' = 0.00081312$. Step $= 0.06$, giving $h = 28.20$.
Step 3: the expected total is now 54.000, so the step is 0 and we stop.
$\boxed{\hat{h} = 28.2 \text{ rating points}}$. Newton converges this fast in practice: our program needs 3 to 5 steps per fit.

4. How sure are we: standard error and confidence interval

4.1 Fisher information

How sharply the log-likelihood peaks tells us how precisely the data pins down h. The *observed Fisher information* is the curvature at the maximum,

$$I(\hat{h}) = -\ell''(\hat{h}) = K^2 \sum_i E_i(1 - E_i)$$

A standard result for maximum-likelihood estimates is that, for large samples, $\hat{h}$ is approximately normal around the true h with variance $1/I$:[1]

$$\hat{h} \;\approx\; \mathcal{N}\!\left(h,\; \frac{1}{I(\hat{h})}\right), \qquad \mathrm{SE}(\hat{h}) = \frac{1}{\sqrt{I(\hat{h})}}$$

4.2 The 95% confidence interval

Since 95% of a normal distribution lies within 1.96 standard deviations of its mean,

$$\hat{h} \pm 1.96\,\mathrm{SE}(\hat{h})$$

is an approximate 95% confidence interval: if we repeated the study many times, about 95% of such intervals would contain the true h. We use the normal distribution and not a t-distribution

[1] The term $E(1 - E)$ in I is the variance of a game that is either won or lost. Draws make results less spread out, so the true variance is a bit smaller and our intervals are slightly conservative (a little too wide). That errs on the safe side.

because the variance comes from the model, not from a small sample estimate, and with millions of games the two would be identical anyway.

Worked example: the confidence interval for the 100 games

At $\hat{h} = 28.2$: $I = 0.00081310$, so SE $= 1/\sqrt{0.00081310} = 35.07$ points. The 95% CI is $28.2 \pm 1.96 \times 35.07 = 28.2 \pm 68.7$, that is from -40.5 to $+96.9$.
With only 100 games we can't even say whether White has an advantage: the interval includes 0. Each game adds at most $K^2/4$ to I, so the SE shrinks like $1/\sqrt{N}$. With N games, SE $\geq 2/(K\sqrt{N}) \approx 347/\sqrt{N}$ points: about 35 for 100 games, 0.35 for a million, and a few hundredths of a point for our 160 million.

5. Aggressive vs drawish: the z-test for a difference

5.1 The test

We fit one h for all games in the four aggressive openings, $\hat{h}_A$ with standard error SE_A, and one for the four drawish openings, $\hat{h}_D$ with SE_D. They come from different games, so they are independent, and for independent quantities variances add:

$$\mathrm{Var}(\hat{h}_A - \hat{h}_D) = \mathrm{SE}_A^2 + \mathrm{SE}_D^2, \qquad \mathrm{SE}(\hat{h}_A - \hat{h}_D) = \sqrt{\mathrm{SE}_A^2 + \mathrm{SE}_D^2}$$

If the openings made no difference, the true difference would be 0, and

$$z = \frac{\hat{h}_A - \hat{h}_D}{\sqrt{\mathrm{SE}_A^2 + \mathrm{SE}_D^2}}$$

would be approximately standard normal. The 95% CI for the difference is

$$(\hat{h}_A - \hat{h}_D) \pm 1.96\sqrt{\mathrm{SE}_A^2 + \mathrm{SE}_D^2}$$

Worked example: two groups

Suppose the aggressive openings give $\hat{h}_A = 30$ with $\mathrm{SE}_A = 2$, and the drawish openings give $\hat{h}_D = 22$ with $\mathrm{SE}_D = 3$.
Difference: $30 - 22 = 8$ points. Standard error: $\sqrt{2^2 + 3^2} = \sqrt{13} = 3.61$.
$z = 8/3.61 = 2.22$, so $p = 2\,(1 - \Phi(2.22)) = 0.027$, and the 95% CI is $8 \pm 1.96 \times 3.61 = 8 \pm 7.1$, that is from 0.9 to 15.1 points.
Notice that the individual CIs (30 ± 3.9 and 22 ± 5.9) overlap, yet the difference is significant. Overlapping intervals do *not* mean "no difference": always test the difference itself.

The same test for time controls. Each speed (Bullet, Blitz, Rapid, ...) is also fitted as its own set of games, and we compare each speed with the next faster one with exactly this test, for example $z = (\hat{h}_{\text{Blitz}} - \hat{h}_{\text{Bullet}})/\sqrt{\mathrm{SE}_{\text{Blitz}}^2 + \mathrm{SE}_{\text{Bullet}}^2}$.

What this test doesn't do. Gambits are played more at some ratings than others, and h depends on rating. So part of a difference between the groups can come from *who plays them* rather than the openings themselves. The report shows each group's average rating, and a chart comparing the two groups bracket by bracket, so this can be checked by eye.

6. Big data: significance vs size

With 160 million games, standard errors are tiny. A difference of half a rating point can have $p < 10^{-10}$ and still be completely unimportant for anyone playing chess. So in the report:

- every effect is given with its *size* and a 95% CI (in rating points, or percentage points for draw rates), not just a p-value;
- we treat a difference of a few rating points as small, whatever its p-value;
- we read how h changes with rating from the bracket-by-bracket charts.

Assumptions to keep in mind

- **Independence.** The formulas assume games are independent. Active players contribute many games each, so the real uncertainty is somewhat larger than the CIs.
- **The Elo curve.** We assume Lichess ratings predict results through the Elo formula. Lichess uses Glicko-2, whose expected score is the same curve for established players, so this is a good approximation.
- **Large-sample normality.** The CI and z-tests rely on $\hat{h}$ being approximately normal, which holds when each fit uses many games. That's why brackets with fewer than 1,000 games are left out.